QPCODE **Reg.No : .....**

Name : .....

MAHATMA GANDHI UNIVERSITY, KOTTAYAM

MGU-UGP (HONOURS) THEORY EXAMINATION OCTOBER 2025

THIRD SEMESTER

DSE – MG3DSEPHY201 – COMPUTATIONAL PHYSICS- C++ PROGRAMMING

(2024 ADMISSION ONWARDS)

Duration: 1.5 HOURS **Maximum Marks: 50**

Remember(K), Understand(U), Apply(A), Analyse(An), Evaluate(E), Create(C), Skill(S), Interest(I) and Appreciation(Ap)

Students should attempt at least one question from each course outcome to enhance their overall outcome attainability.

Part A

Short Answer Type Questions

Answer any **10** questions

Each question carries **2** marks

1.	Define algorithm with an example.	K / CO1
2.	What are unary operators? Give examples.	U / CO1
3.	What is the function of 'if' statement in C++?	U / CO2
4.	Differentiate between while and do-while loops.	U / CO1
5.	Define an array	K / CO1
6.	Define a function	K / CO1
7.	Write the syntax for function definition.	K / CO2
8.	How to access array elements?	U / CO2
9.	How can we call a function?	U / CO2
10.	What are pointers?	K / CO3
11.	What are member functions?	U / CO3
12.	What is polymorphism?	K / CO3

13	What are array of objects?	U / CO3
14.	What is function overloading?	U / CO3

[2x10 = 20]

Part B

Short Essay Type Questions

Answer any **6** questions

Each question carries **5** marks

15.	Explain the steps involved in developing a C++ program.	U / CO1
16.	Discuss different types of operators in C++ with examples.	U / CO1
17.	Write a program to find the largest of three numbers using if-else.	U / CO2
18.	Describe different types of loops with syntax and example.	U / CO2
19.	Explain the switch statement with syntax and example.	U / CO2
20.	Explain one-dimensional and two-dimensional arrays with examples.	U / CO2
21.	Write the syntax and usage of a function. Differentiate between user-defined and library functions	U / CO2
22.	Define class and object. Explain how to declare and create them in C++.	U / CO3

[5x6 = 30]